

Film Study: All-22, Charting a Game, and Scoring Yourself

Watch it again the way a coach does, turn a game into rows you can count, and find out how good your Sunday calls really were.

Super Bowl LX, February 8, 2026. Seattle sacked Drake Maye 6 times, and every one of them looked the same on television: a pass rusher, a quarterback going down, a replay of the rusher's celebration. What the broadcast never showed was *why*. On 5 of those 6 sacks, the charting says, Seattle rushed only four players, which means New England had at least five blockers for four rushers and seven Seahawks were covering. So was it the rush? The coverage? A quarterback holding the ball? A play call that sent everyone deep on the wrong down? You can't answer that from the couch on Sunday. You can answer it on Monday, with the coaches' film, a pause button and a routine, and that's where this chapter starts. It ends with the other thing Monday is for: taking the season of predictions you've been logging since Part 1 and finding out, with a number, how good they were.

You'll be able to...

- Get the All-22 coaches film for a game, set up to study it, and run the four-pass film-study protocol on any snap: freeze at the snap and check your read, then watch the line, the quarterback, the coverage and the skill players, one pass each, and write one line saying why the play worked or failed.
- Chart a full game with a fixed template and codebook, starting from a skeleton built automatically from nflverse play-by-play, and turn the finished chart into tendencies.
- Score a season's prediction log in Python: hit rate, Brier score, skill against honest baselines, and a calibration plot, and say what each result tells you to change.
- Tell calibration (knowing how sure to be) apart from resolution (knowing more), and know why a reader needs both.
- Know where to keep learning once the course is over: coaching clinics, the film-breakdown and analytics communities, and the Big Data Bowl.

i You'll need

This is the last chapter, and it leans on three earlier ones more than the rest:

- **The live routine** (Watching Live: The Master Checklist and Anticipation): the master pre-snap checklist (situation, personnel, formation, motion response, box, shell, leverage, pressure tells, call) and the stated confidence you put on every call, logged in the final form of the prediction log.
- **Charting data** (Tendencies: PROE, Personnel, Motion, and Coverage from Charting Data): xpass is the chance of a pass from the situation alone; coverage charting labels each dropback's coverage after the snap, by a person watching film; and every charted label is a judgement with known caveats.
- **The film itself** (How to Watch a Broadcast Like You Mean It): coaches film shows every snap from a high sideline angle (the All-22) and from a high end-zone camera, with no replays or graphics.

What Sunday can't tell you

Live watching, done well, is prediction. You read the situation and the formation, make a call with a number on it, and the snap tells you whether the call was right. But the snap only answers the first layer. It tells you *pass*; it doesn't tell you whether the coverage you called was the coverage that was played, because the camera was on the quarterback when the safeties turned. It tells you *sack*; it doesn't tell you whose fault the sack was. It tells you *seven-yard run*; it doesn't tell you whether the run went where it was designed to go or bounced somewhere else because a block failed. And after a season of logging, a nagging question: you've been writing "70%" next to calls for months. Do your 70% calls come true seven times in ten?

Those are three different gaps, and this chapter closes each with its own tool:

1. **The film** answers *what happened*, for every player, on every snap. It's the only source that shows all 22 at once for the whole play.
2. **The chart** turns what happened into rows: one per snap, the same fields every time, so a game can be counted instead of remembered. It is also the answer key for your log's second layer.

3. **The score** turns your log into a number that rewards both being right and being honest about how sure you were, and a plot that tells you which way to adjust.

Coaches have done the first two for as long as there has been film. In an NFL building, Monday morning belongs to the last game: every player graded on every snap, the staff deciding what to fix, and the quality control coaches' tags on every snap feeding the reports the coordinators read (The Game Plan: A Week Inside an NFL Building). The third is borrowed from weather forecasters, who have been scored this way since 1950. You'll do a fan's version of all three, with free data, an affordable film subscription and a few dozen lines of Python.

! Why it exists: watching the same play four times

Why not just watch the replay carefully once? Because attention is narrower than it feels. In a famous experiment, viewers asked to count basketball passes in a short video missed a person in a gorilla suit walking through the middle of the scene; about half of them never saw it.¹ Football film is that experiment on every snap: watch the quarterback and you won't see the safety rotate; watch the safety and you won't see which lineman lost. Coaches solved this long ago by dividing the film: the offensive line coach watches the line, the secondary coach the secondary. A fan has no staff, so you divide the film by *time* instead, watching the same snap several times, a different group each time.

What it gives up: time. Four passes on every snap of a game is hours of work, which is why the protocol below comes with a short version.

Getting the film

You met coaches film in How to Watch a Broadcast: every snap shown twice, once from the high wide sideline angle that keeps all 22 players in frame, and once from a high camera behind the end zone. Fans have been able to buy it since 2012; it currently comes with the league's NFL+ Premium subscription, which offered it for the 2026 season and, when the league described the package in 2024, kept an archive back to the 2022 season.² Coaches film isn't posted the moment the game ends; expect it within a day or two. Subscription packages change often, so check what's included before you pay.

If you don't have it, you can still do a lot with what's free. The broadcast's own replays include the skycam and end-zone angles on many snaps. Teams post highlight clips, and several film-breakdown writers post All-22 clips of the plays they analyse (there's a list at the end of the chapter). None of it is every snap from every angle, which is what charting a whole game needs, but it's plenty for studying the snaps that mattered.

Setting up. Three things, side by side if you have the screen for it:

- **The film**, in a player that can pause cleanly on a single frame, step a frame at a time, and slow down to half speed. Most of what you'll do is pause on the snap, play a second and a half, and go back.
- **The chart**, a spreadsheet with one row per snap (the template is below), already filled with everything the play-by-play knows, so you only type what the film adds.
- **Your live log**, the sheet you kept on Sunday. You'll check it against the film, but you won't change it: it's a record of what you knew *before* the snap, and correcting it after watching the film would make your score meaningless.

The game clock in the film is your index. Each row of the play-by-play carries the quarter and the clock at the snap (`qtr` and `time` in `nflverse`), so you can always find the snap you're looking for and keep the film, the chart and the log lined up on the same row.

Pass	Best angle	Why that angle
Freeze at the snap	All-22 (sideline)	The only angle with all 22 in frame: personnel, formation, safeties and corners together
1 · The line	End zone	Looks straight down the line: who stepped where, which gap opened, who came free

¹Daniel J. Simons and Christopher F. Chabris, "Gorillas in our midst: sustained inattention blindness for dynamic events," *Perception* 28 (1999): 1059–1074. In the best-known condition, about half of the observers counting passes failed to notice the person in the gorilla suit.

²The history and the current package are covered, with sources, in How to Watch a Broadcast: Buffalo Bills, "Watch film like a coach with Game Rewind" (July 30, 2012), <https://www.buffalobills.com/news/watch-film-like-a-coach-with-game-rewind-7721721> ; NFL.com, "Launch of NFL Pro highlights third season of NFL+" (2024), <https://www.nfl.com/news/launch-of-nfl-pro-highlights-third-season-of-nfl> .

Pass	Best angle	Why that angle
2 · The quarterback	End zone, then All-22	His feet and eyes from behind; then, from the side, where his eyes were pointing
3 · The coverage	All-22	The safeties' first steps and the corners' technique are only in the wide shot
4 · The skill players	All-22	Routes against the coverage and who was open when the quarterback looked

i Madden vs. real life

In *Madden* the film study is done for you. You picked the play, so you know the call; the play art shows every assignment; and the replay camera can fly anywhere. Real film gives you none of that. You never see the call sheet, so every assignment on the film is something you *infer*: a cornerback who turned and ran with a receiver might be in man coverage, or in a zone that turns into man when the receiver goes deep, or simply beaten. Film study is the craft of making those inferences carefully, writing down how sure you are, and noticing when the next snap from the same look contradicts you. It's the same habit as your live log, applied to the past.

The four-pass film protocol

Here is the routine. It's the course's **film-study protocol**: a fixed way to study each snap on film, the same way every time. You freeze the picture at the snap and check your pre-snap read; then you watch the play four times, once for each group (the line, the quarterback, the coverage, the receivers and backs); then you write one line saying why the play worked or failed.

The four-pass film protocol

one snap, about two minutes

WATCH FOR	ANGLE	WHY HERE
0 Freeze at the snap Pause on the snap. Check the log's pre-snap columns: personnel, formation and strength, motion and the defense's answer, box, shell, leverage. Write what you missed live.	All-22	<i>Your read was made here; grade it before you know the result</i>
1 The line (and the front) First step: forward (run) or back (pass). Run: zone or gap, who pulled, who was left unblocked on purpose, where the hole opened. Pass: blockers against rushers, the slide, who came free, and when.	End zone	<i>The line decides how much time everyone else gets</i>
2 The quarterback Drop and launch point; where his eyes went first; on time or late (count it); the throw he chose against the one that was there; what pressure did to him. Runs: the mesh and any read.	End zone, then All-22	<i>Judge what he did with the time pass 1 measured</i>
3 The coverage Safeties first: their first step, where they finished (middle open or closed). Corners: bail, squat or trail. Who traveled with motion, who blitzed, who replaced him. Name it, with a confidence.	All-22	<i>The hardest read; by now you know where the ball went</i>
4 The skill players The routes as one concept; who was open at the moment the quarterback looked; separation and yards after the catch. Runs: the back's aim, his cut, yards after contact; blocks by receivers and tight ends.	All-22	<i>Routes only make sense against the coverage</i>
✓ The 'because' line One line: why it worked or failed, and whose play it was (the call, the line, the quarterback, the coverage, the skill players). Then fill in the chart row.		<i>A snap you can't explain in one line needs another pass</i>

Figure 1: The four-pass film protocol on one card. Start frozen at the snap (step 0) and check the pre-snap columns of your log against the film. Then watch the play four times, a different group each time: the line, the quarterback, the coverage, the skill players. The order runs from what sets the play's clock (the line) to what uses it (the routes and the run). Finish with one 'because' line that names whose play it was. The middle column says which angle of the coaches film to use, the right-hand column why the step sits where it does.

Each step deserves its reasons.

Step 0: freeze at the snap

Before you watch anything move, pause on the frame where the ball is snapped and grade your Sunday read. The pre-snap columns of your log (personnel, alignment, box, shell) were your best guesses from a tight broadcast shot; the All-22 shows the truth. Was it really two-high, or was the second safety already at eight yards and creeping? Was the box six or seven? Which way did the defense set its strength, and did a linebacker shift when the motion went? Write what you missed in a notes column. Don't overwrite the log.

This step does two jobs. It grades the *inputs* of your call, which your score can't: a 70% pass call that came true is a good result, but if you'd miscounted the backfield it was a lucky one. And it trains your live eye. After a few games you'll find you're consistently

wrong about one thing (a safety's depth is a common one, because the broadcast angle flattens the field), and you'll start correcting for it on Sunday.

Pass 1: the line

Use the end-zone angle and watch only the ten big men: the five offensive linemen and whoever lines up across from them, plus any back or tight end who blocks. The first thing to see happens in the first half-second: **did the line step forward or back?** Linemen who fire out are blocking a run; linemen who sit back are protecting a pass (and on play-action they'll try to fake the first, so look at how far they go).

On a run, ask what scheme it was. If every lineman steps the same direction together, it's zone blocking (Zone Running). If linemen down-block toward the middle while someone pulls across behind them, it's a gap scheme (Gap and Power Runs). Then find the defender nobody blocked. On most runs one is left alone on purpose (the backside end on power, the read key on an option), and knowing which one tells you what the play was meant to do. Finally, where did the hole open, and was that where the back ran?

On a pass, count. How many rushed, how many blocked? Which way did the line slide, and did a back or tight end stay in? Then the most useful habit in pass-protection film: **start counting when the ball is snapped**. If a rusher gets to the quarterback in under two and a half seconds, the protection lost. If the quarterback still has the ball at three and a half seconds, something else went wrong first, and pass 2 and pass 3 will tell you what. (Those thresholds are rules of thumb, not laws; a seven-step drop and a quick game have different clocks, so judge against the drop.)

Why first: the line sets the clock for everyone else. A quarterback with 1.8 seconds and one with 3.5 seconds are playing different games, and a run's whole design is visible in the line's first two steps.

Pass 2: the quarterback

Now watch only the quarterback, starting from the end-zone angle. His feet: a quick drop from the shotgun, a five- or seven-step drop from under center, a run fake, a rollout. His launch point, the spot he throws from. His eyes: where does he look first, and does he move them? A quarterback working a progression turns his head from one receiver to the next in order; one throwing on rhythm hitches once and lets it go at the top of his drop. Count when the ball comes out, and switch to the All-22 to see what he was looking at.

The question that matters most is a comparison: **the throw he made against the throw that was there**. You'll only be able to answer it fully after pass 4, but start it here, at the moment he chose. And on runs, the quarterback still has a job: the mesh with the back, any read he made (on a zone read, did he keep or give, and was it right?), and whether his fake sold anything.

Why second: you can only judge what he did with the time you measured in pass 1. Sacks that were the protection's fault get blamed on quarterbacks every week, and the reverse.

Pass 3: the coverage

Switch to the All-22 and start at the snap with the **safeties**. Their first step tells you more than anything else in the secondary: a safety who backpedals straight is playing his half or his quarter; one who runs toward the middle is becoming the single high safety; one who drops down toward the line is rotating into a short zone or taking a man. A second later, ask the question from the coverage chapters: is the middle of the field closed (one deep safety in the middle) or open (two deep, the middle between them)? That single answer splits the coverages into families (Disguise, Rotation, and Split-Field Coverage).

Then the corners: did each one **bail** (turn and run to a deep third), **squat** (sit on a short zone near the sideline), or **trail** (follow his receiver underneath, with help over the top)? Who traveled with the motion before the snap? Who blitzed, and who dropped into the spot he left? Defenders looking at the quarterback are usually in zone; defenders looking at a receiver are usually in man. Then write the coverage family, and a confidence. Coverage read from film is still an inference, as Tendencies warned about professional charters' labels: match coverages turn from zone into man by rule, and a busted assignment can look like a design.

Why third: coverage is the hardest read on the film and deserves a clean pass with nothing else to look at. And by now you know where the ball went, so you can tell whether the coverage was beaten, or whether the ball went somewhere the coverage had already given away.

Pass 4: the skill players

Last, the receivers, tight ends and backs. On a pass, name the routes as one concept (Concepts versus Coverages): smash, flood, mesh, a four-verticals look. Then go back to the moment pass 2 found, when the quarterback looked or threw, and ask **who was open then**. "Open" at the moment of the throw is the only kind that matters; a receiver who comes open a second after the ball is

gone wasn't an option. Notice separation, how the receiver beat his man or found the hole in the zone, and what happened after the catch.

On a run, watch the back: where he aimed, where he cut, how many yards he gained before anyone touched him and how many after. Then the blockers in space: did a receiver's crack block seal the edge, did the tight end reach his man?

Why last: routes only make sense against the coverage you identified in pass 3, and "open" only makes sense at the moment you found in pass 2.

The "because" line

Finish every snap with one line that says why it worked or failed, and whose play it was. There are five candidates, and a disciplined charter picks one first and adds a second only if he must:

- **The call.** The play was wrong for the defense it met: a run into an eight-man box, a deep concept against a coverage built to stop it. Nobody failed; the play-caller lost the guess.
- **The line.** A block lost and the play died there.
- **The quarterback.** He had time and an open man, and he did something else.
- **The coverage.** On defense: a rotation took the throw away, or a defender lost his man.
- **The skill players.** A drop, a missed cut, a broken tackle, or a great catch.

"Pass, incomplete, because the boundary safety rotated down and took away the out; QB stayed with it too long" is a good line. "Bad play" isn't. If you can't write the line, you haven't seen the play yet: watch it again.

Why this order, and why not more passes

The order runs from **what sets the clock to what uses it**. The line decides how long the play lasts; the quarterback decides what to do inside that time; the coverage decides what's available; the skill players decide whether it gets used. Watch it in that order and every pass gives you context for the next one. Watch the receivers first and you'll call a receiver "open" who the quarterback never had time to look at.

Why four, and not one per position? Because a fan's eye can hold about one group at a time, and these four groups answer different questions. More passes add little: by pass 4 you've seen every player, and what's left is re-watching the moment that matters, which you should do as often as you need. A snap takes about two minutes this way once you're practised, more at first.

A full game, in the time you have

About 60 to 65 offensive snaps a side, at two minutes each, is more than four hours for both teams. Few people have that on a Monday. So run the protocol in layers:

- **Every snap:** step 0 (freeze and check the pre-snap columns) and pass 3 (the coverage), because those are the two things the broadcast hid from you and your log's second layer needs.
- **Every third down, red-zone snap, sack, turnover and explosive play** (a run of 10 yards or more, a pass of 20 or more): the full protocol. These are the snaps that decided the game, and there are usually 20 to 30 of them.
- **Everything else:** at real speed once, then the chart row.

That's two or three hours for one team's offense against the other's defense, which is a realistic weekly habit.

Film room: one snap, four passes

i Film room: Seahawks 2025 — Super Bowl LX, the strip-sack

The game: Seattle beat New England 29–13 in Super Bowl LX on February 8, 2026, at Levi’s Stadium in Santa Clara.³ Late in the third quarter Seattle led only 12–0, all on field goals, and New England had the ball.

The snap: third-and-6 at the New England 44, :16 left in the third quarter. The charting records the Patriots in 11 personnel in the shotgun with one back and motion before the snap; Seattle in its dime package (six defensive backs) with 6 in the box; **4 rushers**; and coverage labelled **2-Man**: man coverage underneath with two safeties deep (2-Man). Derick Hall sacked Drake Maye, forced a fumble, and Byron Murphy II recovered at the New England 37. Five snaps later Seattle scored its first touchdown, a 16-yard pass to AJ Barner, and led 19–0.⁴

Why this snap: it’s the question from the top of the chapter. A four-man rush got the sack; the film decides whether the rush won it or the coverage did.

The figure is a reconstruction in the same format as the course’s other replays: what was charted is exact, and the rest (who lined up where, the routes, which tackle Hall beat and how long it took) is a typical picture of that call, drawn to show what each pass of the protocol looks at. On the real film, you’d check every one of those details yourself.

³Score, date and site: nflverse schedules and play-by-play (game [2025_22_SEA_NE](#)); “Super Bowl LX,” Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_LX. Quarter-by-quarter, Seattle led 3–0, 9–0 and 12–0 at the ends of the first three quarters, all on Jason Myers field goals.

⁴nflverse play-by-play, FTN charting and participation for game [2025_22_SEA_NE](#), the Patriots’ third-quarter play with 0:16 left: “D.Maye sacked at NE 37 for –7 yards (58-D.Hall). FUMBLES (58-D.Hall) ... RECOVERED by SEA-91-B.Murphy at NE 37.” Personnel, alignment, motion, box, rushers and the coverage label ([2_MAN](#)) are FTN’s charting, not the teams’ calls. The touchdown that followed: “S.Darnold pass deep right to 88-A.Barner for 16 yards, TOUCHDOWN,” 13:29 of the fourth quarter.

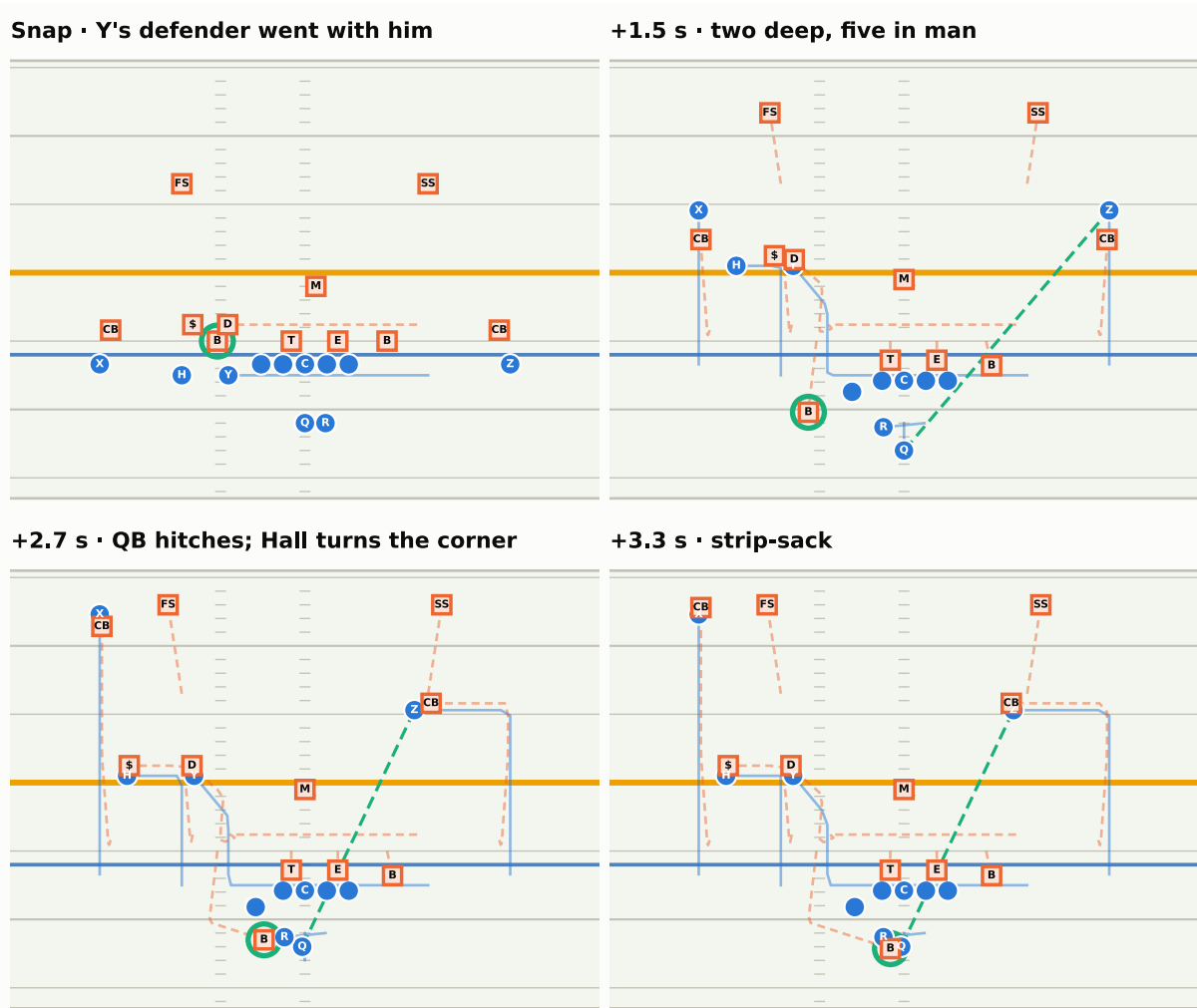


Figure 2: The strip-sack, third-and-6 (a reconstruction, simulated in Big Data Bowl format). Charted facts: 11 personnel in the shotgun with one back and motion; Seattle's dime (six defensive backs), six in the box; four rushers; 2-Man (man coverage underneath, two safeties deep); a sack by Derick Hall (B, ringed), who forced a fumble. Illustrative: the alignments, the motion (here the tight end, Y, across to the left, with his defender, D, following him), the routes, the protection, and the time Hall took. What to notice: the second panel, where every receiver has a defender on his inside hip and a safety above him, and the third, where the quarterback has hitched up with nowhere to go while Hall turns the corner on the left tackle.

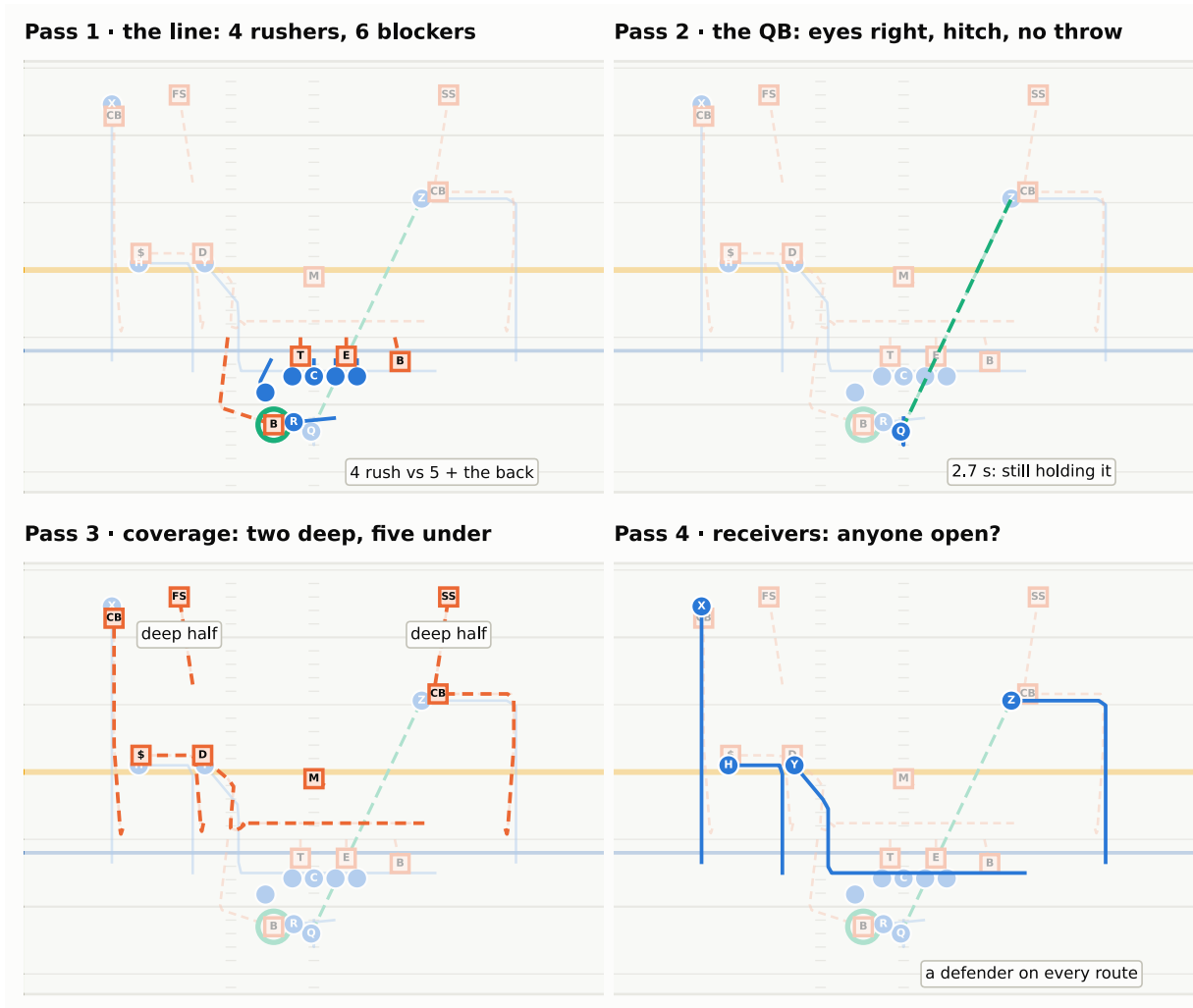


Figure 3: The same snap at 2.7 seconds, watched four times. Each panel fades everyone except the group that pass watches, with the paths they have run so far. Pass 1, the line: four rushers against five linemen and the back; three rushers are stalemated, and Hall (ringed) has run around the left tackle. Pass 2, the quarterback: a short drop, eyes to the right (the green dashed line), then a hitch up into the pocket. Pass 3, the coverage: both safeties in deep halves, the four man defenders trailing on the receivers' inside hips, and the middle linebacker (M) sitting in the short middle because the back stayed in to block. Pass 4, the skill players: four routes, each with a defender attached and a safety above. A reconstruction, as in the previous figure.

Now run the protocol on it, the way you would on the real film.

Step 0, the freeze. Third-and-6 near midfield, down 12, sixteen seconds left in the third quarter: a pass, near certain. The offense is in 11 personnel in the shotgun; the defense in dime with six in the box and two safeties at about 12 yards. The motion is the useful clue: when the tight end crossed the formation, his defender went *with* him. That's the man/zone indicator from Motion and Shifts saying man, and the two deep safeties say two-high. Put them together and the pre-snap read is 2-Man or a disguised rotation; on third-and-6 2-Man is a common answer, because it puts a defender on every receiver short of the sticks and two safeties over the top of them.

Pass 1, the line. Four rushers against six blockers (five linemen and the back). The line had the numbers, so if the sack came fast, someone lost a one-on-one block. Count from the snap: in the reconstruction, Hall gets to Maye at about 3.3 seconds, which on a shotgun drop is *late*. The charting doesn't record that time; on the real film it's the first thing you'd measure, because it decides how much of the sack is the left tackle's.

Pass 2, the quarterback. Maye takes his drop, looks right, hitches up into the pocket and still doesn't throw. That's a quarterback with time and no answer, or one who missed an answer. Which, pass 4 will say.

Pass 3, the coverage. Both safeties backpedal into deep halves. Underneath, every receiver has a defender on his inside hip in trail technique, so anything short and inside is contested and anything deep has a safety coming. The middle linebacker has the back, and when the back stays in to block, he sits in the short middle as an extra defender, taking away the throw a quarterback looks

for against man: the crossing route underneath. Seven defenders in coverage against four receivers is the whole idea of rushing four: the rush doesn't have to win quickly, because the coverage is designed to hold out.

Pass 4, the skill players. Against 2-Man, receivers need to win on their own: the route that beats it is the one that gets *away* from a trailing defender toward the sideline or back toward the quarterback, before the safety arrives. At the moment Maye hitched, nobody had. That's the question to answer on the real film, frame by frame: was there a window at 2.5 seconds he passed up?

The "because" line. In this version: *Sack because of the coverage: 2-Man took away everything inside and short of the sticks, and the four-man rush won late, at 3.3 seconds.* If on the real film Hall had arrived in two seconds, it would read *because the left tackle lost*, and if a receiver had been open at 2.5 seconds, *because the quarterback held it.* Same play in the box score, three different lessons. That's what the film is for.

Common misconception

"A sack is a pass-protection failure." Often it isn't. A sack is credited to the rusher who gets there, and the offensive line usually takes the blame, but a sack is the end of a chain: protection, quarterback, coverage, routes. A defense that rushes four and covers with seven is betting that its coverage will hold long enough for four rushers to win *eventually*, and when that bet pays off it's a coverage sack. In Super Bowl LX Seattle rushed four on 45 of New England's 53 dropbacks and got 5 of its 6 sacks that way. You can't tell from the box score which of those were the line's fault and which were the coverage's. The film can.

Charting a game

The film tells you what happened on one snap. To learn what a team *does*, you need every snap in the same form, and that's charting. **Game charting** means recording every snap of a game as one row of the same fixed fields (situation, personnel, formation, motion, the defense's look, what happened after the snap, the result), coded the same way every time, so a game becomes data you can count. Everyone calls it charting for short.

You've been using other people's charting since Part 13. When the participation file says a snap was Cover 6 against 11 personnel, somebody watched that snap and wrote those codes. NFL staffs do it too: their quality control coaches tag every snap of every opponent, and the tags become the tendency report (The Game Plan). Charting your own game puts you in that chair.

Why chart, when FTN and nflverse already do? Three reasons.

1. **Timing.** The public charting doesn't all arrive at once. The play-by-play for a game is usually available within a day or so; FTN's charting (alignment, motion, play-action, rushers) is updated weekly during the season; but personnel and coverage labels, in the participation file, arrive only after the season ends, around February.⁵ If you want to know what coverage Seattle played against 12 personnel in *this* season, the public data can't tell you until the season is over. Your chart can tell you tonight.
2. **What nobody publishes.** The public files don't record the pre-snap shell (only the coverage after the snap), the corners' leverage, the formation's strength, which way the run went relative to that strength, the run scheme, the pass concept, the routes of receivers who weren't targeted, or who lost the block on a sack. Those are exactly the fields that make a tendency useful, and exactly what step 0 and the four passes record.
3. **The answer key.** Your live log's second layer (the coverage family, the rush, the direction of the run) needs something to be scored against. On a 2026 game, until the participation file arrives, your chart is the only answer key there is.

The charting sheet

Here is the template. One row per snap, in seven blocks of columns:

⁵Release timing from the nflverse data dictionaries and release notes: FTN charting is updated weekly during the season and participation (personnel and coverage) is released after the season ends; https://nflreadr.nflverse.com/articles/dictionary_ftn_charting.html , https://nflreadr.nflverse.com/articles/dictionary_participation.html . Play-by-play for the current season is updated shortly after games.

FROM PLAY-BY-PLAY					OFFENSE BEFORE THE SNAP					DEFENSE BEFORE THE SNAP			
#	qtr clock	dn&to	ball on	hash	pers	align	form	str	mot	dpers	box	shell	CB
1	Q1 11:52	1&10	NE 28	M	22	U1			M	nickel	7		
2	Q1 11:21	2&4	NE 34	L	12	U1			-	nickel	7		
3	Q1 10:40	3&2	NE 36	M	11	S1			M	nickel	7		
4	Q1 10:04	1&10	NE 43	L	11	U1			M	nickel	6		
5	Q1 09:28	2&2	SEA 49	L	22	U2			-	nickel	7		
6	Q1 08:51	1&10	SEA 45	R	11	S1			-	nickel	6		
7	Q1 08:09	2&20	NE 45	R	11	S1			M	dime	6		
8	Q1 07:23	3&9	SEA 44	R	11	S1			M	dime	5		

AFTER THE SNAP				THE PLAY AND RESULT					YOUR LIVE LOG				FILM
#	rush	cov	rot	play	concept	dir	yds	succ	call	conf	l2	l2c	because
1	-	-		R		M	+6	✓	R	.65	2-hi	.65	
2	-	-		R		M	+2	✗	R	.70	1-hi	.50	
3	4	C9		P		L	+7	✓	P	.80	1-hi	.55	
4	4	C2		PA		L	+8	✓	R	.65	2-hi	.65	
5	-	-		R		R	+4	✓	R	.85	1-hi	.50	
6	4	C2		P		-	sk -10	✗	P	.60	2-hi	.65	
7	4	C2		P		-	+11	✓	P	.80	1-hi	.50	
8	6	C0		P		R	+0	✗	P	.90	1-hi	.55	

Figure 4: The charting sheet, with New England's first eight snaps of Super Bowl LX filled in from public data. The sheet is one long row per snap, printed here in two tiers that share the snap number. Grey blocks fill themselves from the play-by-play; blue and orange blocks (the offense's and the defense's look) and the after-the-snap block are what you chart from film, filled here from FTN's charting and the participation file where those record the field; the cream-shaded cells are fields no public source records (formation, strength, the pre-snap shell, the corners, the rotation, the play concept, the 'because' line), which only your own film can fill. The green block is your live log, filled here with calls made by rule from the pre-snap columns (as in Watching Live), standing in for yours; its second layer is a call on whether Seattle would play two-high, at Seattle's 2025 rate for the down.

Abbreviations are in the codebook below.

And here is the codebook that goes with it. The codes matter less than the rule behind them: **decide them before you start, write them on the sheet, and use them the same way on every snap.** A chart in which "Cover 3" meant one thing in the first quarter and another in the fourth can't be counted.

Table 1: The codebook for the charting sheet. ? is always allowed and is better than a guess.

Column	Codes	The rule
<code>pers</code>	11, 12, 21, 13, 10, 22 ...	Backs, then tight ends; count as they run on
<code>align</code>	U, S or P plus the number of backs: U1, S0, P1	Where the quarterback stands, how many in the backfield
<code>form</code>	2x2, 3x1, bunch, empty 3x2, I, offset I ...	Pick one name per look and reuse it
<code>str</code>	L, R	The strong side, by one rule you write at the top of the sheet (tight end, receivers, or field)
<code>mot</code>	–, M (motion), S (shift), J (still moving at the snap)	What moved, and when
<code>dpers</code>	base, nickel, dime	Count the defensive backs
<code>box</code>	a number	Defenders inside the tight ends, within about five yards of the ball
<code>shell</code>	0, 1, 2	Deep safeties at the snap (before any rotation)
<code>CB</code>	P or O, plus i or o	Press or off; inside or outside leverage
<code>rush</code>	a number	Rushers after the snap
<code>cov</code>	C0, C1, C2, 2M, C3, C4, C6, C9, ?	The coverage played after the snap
<code>rot</code>	–, B, F, L	Rotation: none, to the boundary, to the field, late
<code>play</code>	R, P, PA, SCR, RPO	Run, dropback, play-action, screen, run-pass option
<code>concept</code>	IZ, OZ, PWR, CTR, DUO; smash, flood, mesh, stick ...	The run scheme or pass concept, in the course's names
<code>dir</code>	L, M, R	Where the run hit or the pass went
<code>because</code>	free text	One line from the protocol: whose play it was

Three rules keep a chart honest:

- **? is a code.** If the broadcast cut away, or you can't tell Cover 1 from Cover 3 on a late rotation, write ?. A guessed label looks exactly like a real one in the totals and quietly poisons them.
- **Code what you see, not what you expect.** Charting a team you've studied, you'll be tempted to write its favourite coverage before you've watched the safeties. That's how a tendency becomes a self-fulfilling one in your chart.
- **Never edit the live log.** The chart is what happened; the log is what you knew before it happened. Keep them in separate columns and score one against the other.

The skeleton: let the play-by-play do the typing

A third of the sheet is already known before you press play: the snap order, the clock, down and distance, the ball's spot, the result. nflverse publishes all of it within a day or so of each game, so the sensible way to start a chart is to build the skeleton in code and spend your film time only on what the film adds. This function does it for any game in any season nflverse covers:

```
import pandas as pd
from gridiron.data import pbp

FILM_COLUMNS = ["pers", "align", "form", "str", "mot", "dpers", "box", "shell", "CB",
                "rush", "cov", "rot", "concept", "because"]

def chart_skeleton(game_id, team):
    """One row per snap of `team`'s offense, with the play-by-play columns filled and the film columns empty."""
    season = int(game_id[:4])
```

```
d = pbp([season])
d = d[(d.game_id == game_id) & (d.posteam == team) & d.play_type.isin(["pass", "run"])]
sheet = pd.DataFrame({
    "play_id": d.play_id.astype(int), "qtr": d.qtr.astype(int), "clock": d.time,
    "down": d.down, "togo": d.ydstogo, "ball_on": d.yrdln,
    "play": d.play_type.map({"pass": "P", "run": "R"}),
    "dir": d.pass_location.fillna(d.run_location).str[0].str.upper(),
    "yds": d.yards_gained, "success": d.success, "epa": d.epa.round(2)})
for c in FILM_COLUMNS:
    sheet[c] = "" # yours to fill from the film
return sheet.reset_index(drop=True)

sheet = chart_skeleton("2025_22_SEA_NE", "NE") # Super Bowl LX, New England's offense
print(f"{len(sheet)} snaps; first five of the play-by-play half:")
print(sheet.iloc[:5, :11].to_string(index=False))
```

```
67 snaps; first five of the play-by-play half:
play_id  qtr clock  down  togo ball_on play dir  yds  success  epa
279      1 11:52   1.0  10.0  NE 28   R  M  6.0      1.0  0.28
301      1 11:21   2.0   4.0  NE 34   R  M  2.0      0.0 -0.37
323      1 10:40   3.0   2.0  NE 36   P  L  7.0      1.0  1.15
353      1 10:04   1.0  10.0  NE 43   P  L  8.0      1.0  0.58
378      1 09:28   2.0   2.0  SEA 49   R  R  4.0      1.0  0.25
```

Save it with `sheet.to_csv("sb60_ne.csv", index=False)`, open it in a spreadsheet next to the film, and fill the empty columns. (Penalties that wipe out a snap, kneels and spikes aren't in this skeleton; add a row by hand if a wiped-out snap told you something.) For a current-season game, the same call gives you the skeleton the day after the game, weeks or months before the public coverage labels exist.

Is your chart any good?

Professional charters disagree with each other, and you'll disagree with yourself. The cheap test is to chart one quarter twice, a week apart, without looking at the first attempt, and compare. For each film column, count how often the two attempts agree. Personnel and the box should agree almost every time; if they don't, your definition is fuzzy (where exactly does "the box" end?). Coverage will agree less often, and that's normal: it's the hardest column, and the place to use ? more freely. If you want a single number that allows for agreement by chance, Cohen's kappa is the standard one.⁶ You don't need it to learn something: the columns where you disagree with yourself are the ones to stop trusting in your totals, and the ones to practise.

Film room: a whole game, charted

i Film room: Seahawks 2025 — Super Bowl LX, charted end to end

The game: Seahawks 29, Patriots 13, Super Bowl LX. In 2025 Seattle's head coach was Mike Macdonald, its offensive coordinator Clint Kubiak and its defensive coordinator Aden Durde; New England's offensive coordinator was Josh McDaniels.⁷ The Game Plan rebuilt, from public charting, the tendency report New England's staff might have read about Seattle's defense before this game. This is the game itself.

The chart: every one of the game's 137 runs and dropbacks, both offenses, in the sheet's codes, filled from the public charting (play-by-play, FTN and the participation file, which covers the playoffs).⁸ The film-only columns are left empty: they're yours.

What to notice: how much of the game's story is visible in the rows, and how much isn't.

⁶Jacob Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement* 20 (1960): 37–46. Kappa compares the agreement you observed with the agreement two coders would reach by chance, given how often each uses each code.

⁷Seattle's and New England's 2025 coordinators as listed for Super Bowl LX: https://en.wikipedia.org/wiki/Super_Bowl_LX and https://en.wikipedia.org/wiki/2026_Seattle_Seahawks_season (Kubiak offensive coordinator, Durde defensive coordinator).

New England offense (Seattle's defense)

Q1 11:52 · NE 28	run +6	run +2	4-C9 +7	4-C2 +8	run +4	4-C2 sk -10	4-C2 +11	6-C0 +0	punt
Q1 6:07 · NE 36	4-C2 +21	run -3	5-C2 -2	6-C6 sk -10	punt				
Q1 :19 · NE 32	run +5	4-C1 +0	run +0	punt					
Q2 11:16 · NE 35	run +3	4-C1 sk -10	4-C6 +5	punt					
Q2 4:58 · NE 2	run +3	4-C6 +0	4-C2 +9	run +3	5-CMB +0	6-C4 +0	punt		
Q3 14:54 · NE 32	4-C1 +0	run +7	4-C1 +0	punt					
Q3 9:07 · NE 32	run +9	run +0	4-CMB +0	punt					
Q3 5:10 · NE 27	4-C6 sk -2	4-C9 +8	4-C1 +0	punt					
Q3 1:55 · NE 27	run +3	4-C1 +5	4-C6 +0	4-C9 +4	4-2M sk FUM	turnover			
Q4 13:24 · NE 35	4-C6 +6	4-C2 +24	4-C6 TD	TD					
Q4 10:40 · NE 4	4-C2 +0	5-C3 +13	4-C6 +0	4-C1 +4	4-C1 +16	5-C6 +7	4-2M INT	turnover	
Q4 5:29 · NE 38	4-C2 sk -4	4-C2 +6	4-2M +16	5-C4 INT	pick-six				
Q4 4:27 · NE 35	4-C2 +11	4-C2 +8	4-C2 +5	4-C6 +7	4-C2 +11	4-C6 +26	4-C1 +0	4-2M TD	TD
Q4 1:49 · NE 6	4-C2 +6	4-2M +5	4-C2 +11	4-C6 +9	4-C6 +0	4-C4 +24	end		

Seattle offense (New England's defense)

Q1 15:00 · SEA 35	run +10	run +0	5-C6 +15	5-2M +23	5-C3 +0	run +3	6-2M +0	FG	
Q1 7:11 · SEA 10	5-C3 +0	4-C2 +4	4-C2 +0	punt					
Q1 3:29 · SEA 13	run +2	run +1	5-C9 +12	run -5	5-C6 +0	3-C3 +0	punt		
Q2 14:10 · SEA 24	5-C9 +0	run +30	5-C3 +0	run +29	run -2	4-C4 +0	run -2	FG	
Q2 9:16 · SEA 29	run +9	run +2	6-C3 +0	5-C3 +7	4-C2 +6	run +4	run +1	5-C1 +0	punt
Q2 2:50 · SEA 43	6-C3 +7	run +7	run +1	4-C2 +7	run +5	4-C4 +7	4-C4 +0	4-C4 +0	FG
Q3 14:02 · SEA 9	run +2	6-C1 +16	5-C1 +11	4-C6 +11	run +3	4-C1 +20	run +1	4-C2 +0	7-C4 +0 FG
Q3 7:34 · SEA 18	4-C3 sk -8	4-C9 +4	run +6	punt					
Q3 3:38 · SEA 21	run -2	5-C1 +0	4-C4 +8	punt					
Q3 :10 · NE 37	4-C6 +11	run +1	5-C1 +0	5-C1 +9	6-C1 TD	TD			
Q4 12:22 · SEA 31	5-C2 +11	run -1	7-C0 +0	6-C6 +0	punt				
Q4 8:37 · NE 38	run +14	run +10	run +2	run +1	4-2M +3	FG			
Q4 2:20 · NE 46	run +3	run -6	run +1	punt					

dropback run successful play

Figure 5: Super Bowl LX on one page: every run and dropback, one cell per snap, one row per drive, New England's offense on top and Seattle's below. Blue cells are dropbacks (including sacks and scrambles), amber cells runs; a dark outline marks a successful play (one that gained enough of the yards needed, by nflverse's EPA-based definition). For dropbacks the top line gives the number of rushers and the charted coverage (2M = 2-Man, C6 = Cover 6, and so on; '-' where none was charted); for runs it says 'run'. The bottom line is the result: yards, a sack, an interception (INT), a lost fumble (FUM) or a touchdown (TD). The right-hand label is how the drive ended. What to notice: the Patriots' first nine drives, none longer than eight snaps, eight ending in punts and the ninth in the strip-sack, against a defense rushing four on almost every dropback and playing two-deep behind it; and Seattle's offense, steady rather than explosive, turning four of its first seven drives into field goals. Source: nflverse play-by-play, FTN charting and participation, game 2025_22_SEA_NE.

Now read the chart the way a coach reads a tendency report: count, then ask why.

From the chart	Super Bowl LX	2025 regular season	League, 2025
New England dropbacks with four rushers	45 of 53 (85%)	76%	70%

⁸nflverse play-by-play, FTN charting (via `load_ftn_charting`) and participation (via `load_participation`) for game 2025_22_SEA_NE; plays with `play_type` pass or run and a down recorded, so two-point tries, kneels, spikes and penalties that wiped out a snap are not on the board. Coverage labels and rusher counts are FTN's charting; FTN's data is used under the CC-BY-SA 4.0 licence. Success is nflverse's `success` flag (EPA greater than zero).

From the chart	Super Bowl LX	2025 regular season	League, 2025
Two-high coverage after the snap (of labelled dropbacks)	73% (51 labelled)	52%	43%
Man coverage (of dropbacks charted man or zone)	28%	—	—
Pressure on New England dropbacks	28%	—	—
Sacks; with four rushers	6; 5	—	—
New England snaps that were dropbacks	80% of 66	New England: 60%	—
Seattle snaps that were runs	44% of 71	Seattle: 48%	—
Kenneth Walker III	27 carries, 135 yards	—	—

Four things jump out of those rows.

Seattle did in the biggest game what it did all season, only more so. Rushing four and playing two deep behind it was Seattle's identity (Fangio's Two-High Revolution and Macdonald's Answer), and the Super Bowl chart is that identity turned up: four rushers on 85% of dropbacks and two-high on 73% of the labelled ones. A tendency report told New England this was coming. Knowing it and beating it are different things.

The four-man rush still got home. Pressure on 28% of dropbacks and 6 sacks, 5 of them with four rushers, is the "coverage holds, rush wins late" bet from the strip-sack paying off across a whole game.

The scoreboard shaped the play-calling. New England dropped back on 80% of its snaps, against 60% in its regular season, because it trailed from the first quarter on and spent the fourth quarter chasing three scores. Read that number as a tendency and you'd call New England's offense pass-happy; it was a game state. Seattle, ahead all night, ran on 44% of its snaps (its season rate: 48%), with Kenneth Walker III carrying 27 times. The neutral-situation filter of Tendencies exists for exactly this reason. One game's chart describes that game, not the team.

And the chart is silent where the film isn't. The public rows can't tell you which of New England's two-high looks were *shown* before the snap and which were *played* after it, which receivers Seattle's man defenders were matched on, or who lost the block on each sack. Those are the cream-shaded columns of the sheet, and they're where the answers to "how did Seattle do it?" live. A full chart of this game, with the film columns filled, is a weekend's project and the best one this chapter can offer you.

Common misconception

"The chart is the truth about the game." A chart is a record of *one charter's judgements*, coded consistently. It's far better than memory and far better than the box score, but its coverage column is an inference from film, its box count depends on where you draw the box, and its "because" line is an opinion. Treat a chart the way Tendencies taught you to treat FTN's: strong evidence about habits across many snaps, weaker evidence about any single one.

Scoring yourself

Now the log. Since Downs, Distance, and Field Position you've been calling run or pass before the snap. Since How to Watch a Broadcast you've written a confidence next to each call, and since Watching Live that confidence has been a stated number between 50% and 90%. This section scores it.

To make that concrete, the rest of the chapter uses an example log: **every Panthers offensive snap of the 2025 season**, 1054 calls across 18 games (17 regular-season games and the wild-card loss to the Rams). The situations and the results are real. The *calls* are simulated: a reader who sees roughly what a model using the situation and the formation sees, through some noise, and

⁹The example log uses every Panthers run and dropback of the 2025 season with an xpass value (nflverse play-by-play joined to FTN charting), including the wild-card game. For each snap, a "situation + formation" model (a logistic regression on nflverse's xpass, the quarterback's alignment and the number of backs, learned on the 2024 regular season) gives the chance of a pass; the simulated reader sees that chance on the log-odds scale multiplied by an overconfidence factor (1.5 in the first game, falling to 0.95 by the last) plus random noise, calls whichever side is more likely, and rounds his confidence to the nearest 5%, capped at 95% in the first eight games and 90% after. Results are the real plays. The readers in the third drill use the same method with fixed factors (A: 2.2, B: 0.35).

who starts the season overconfident and settles down as it goes on.⁹ It's a stand-in for your log, built so that its mistakes are the common ones.

Why hit rate isn't enough

The first number is the one you've been computing since Part 1: the share of calls that came true. The example reader was right on 74% of calls. Against the bars from *How to Watch a Broadcast*, that's good: on the same Panthers snaps, "always pass" would have been right 59% of the time, the down-and-distance rule from your first tally 60%, and nflverse's situation-only xpass model 66%.

But hit rate throws away the confidence column, and the confidence column is where most of a reader's skill shows. Two readers can both be right 70% of the time: one who said 70% on every call and meant it, and one who said 95% on half his calls and 55% on the rest and was wrong on a pile of the 95s. The second reader is making a mistake the hit rate can't see. You need a score that rewards being right *and* being honest about how sure you were.

The Brier score

That score is the **Brier score**: the average, over all your calls, of the squared gap between your stated confidence and what happened, counting what happened as 1 if the call came true and 0 if it didn't.

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (c_i - o_i)^2, \quad c_i = \text{stated confidence}, \quad o_i = \begin{cases} 1 & \text{call came true} \\ 0 & \text{it didn't} \end{cases}$$

Lower is better. Three calls make the arithmetic plain:

Call	Confidence	Came true?	Gap	Squared
Pass	0.80	yes (1)	0.20	0.04
Run	0.60	no (0)	0.60	0.36
Pass	0.90	no (0)	0.90	0.81

$$\text{Brier} \quad (0.04 + 0.36 + 0.81) / 3 = 0.40$$

Read the table as a set of incentives. A confident call that comes true costs almost nothing (0.04). A timid call that misses costs a moderate amount (0.36). A confident call that misses costs a lot (0.81). And a reader who says 50% on every snap, never committing to anything, scores exactly 0.25, whatever happens, because every gap is one half. So **0.25 is the "I have nothing" line**: a log that scores worse than 0.25 has done harm by stating its confidences.

Glenn Brier, a meteorologist at the U.S. Weather Bureau, proposed the score in 1950 to grade forecasts like "70% chance of rain", and it's still the standard way to grade probability forecasts in weather, in medicine, and in the forecasting tournaments that taught *Watching Live* its habit of stating numbers.¹⁰ It doesn't matter which way round you write your calls: scoring "pass, 0.70" against whether it was a pass gives the same number as scoring "my call, 0.70" against whether your call came true. Your log's **conf** and **hit** columns are all it needs.

Why squared? A score you can't game

Why square the gap, rather than just averaging it, or awarding points for confidence on the calls you got right? Because of what each rule rewards you for saying. Suppose that on a certain kind of snap, a call comes true 75% of the time, and you know it. What should you say?

¹⁰Glenn W. Brier, "Verification of forecasts expressed in terms of probability," *Monthly Weather Review* 78, no. 1 (1950): 1–3. Brier's original version summed over both outcomes, so it runs from 0 to 2; the version used here (and almost everywhere today) is half of that for a yes-or-no event, running from 0 to 1.

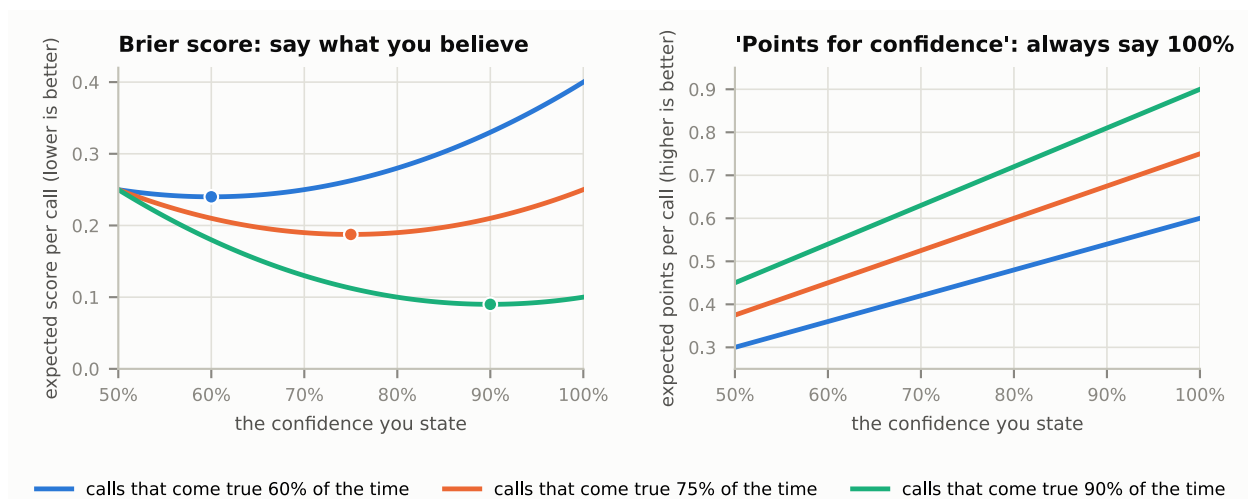


Figure 6: Why the Brier score squares the gap. Each curve is the score you should expect, call after call, if you always state the confidence on the horizontal axis on snaps where your call really comes true 60%, 75% or 90% of the time. Left, the Brier score (lower is better): every curve bottoms out exactly at the true rate (the dots), so the best thing you can do is say what you believe. Right, a naive rule that awards your stated confidence as points when you're right and nothing when you're wrong (higher is better): every line rises to the right edge, so it pays you to say 100% on every call, whatever you believe. A scoring rule with the Brier score's property is called proper.

That property has a name: the Brier score is a **proper** scoring rule. Its expected value is lowest when you state exactly the probability you believe, so there's no way to game it by shading your numbers up or down. It also explains the advice in *Watching Live* to rarely go above 90%: near the bottom of each curve the score is flat, so stating 90% when the truth is 95% costs very little, while stating 99% when the truth is 85% costs a lot.

Skill: compared with what?

A Brier score means little on its own; 0.19 is good or bad depending on how predictable the snaps were. So compare it with forecasters who know less than you, on the same snaps. On the Panthers' 2025 snaps:

Forecaster	What it knows	Brier score
Always 50%	Nothing	0.250
The base rate	The Panthers threw on 59% of snaps; says that every time	0.241
xpass	The situation: down, distance, field position, score, clock	0.205
The example reader	What the reader saw, as logged	0.186
Situation + formation model	The situation, plus the quarterback's alignment and the number of backs	0.175

The **Brier skill score** turns a comparison into one number: one minus the ratio of your score to the reference's. Against the base rate, the example reader's skill is 0.23, meaning the log removed about 23% of the error of a forecaster who only knew the team's pass rate. Against xpass it's 0.09: the reader beat the scorebug, because the formation was working for him. And the last row is the honest ceiling for a reader who only uses those two things. The gap between the reader and that row is the room left: some of it is calibration (stating the right number), some of it is the noise of reading a broadcast in twenty seconds.

That's the comparison to make with your own log. **Beat 0.25** and your confidences are doing more good than harm. **Beat the base rate** and you know more than the team's tendency. **Beat xpass** and you're reading something the scorebug doesn't show.

Calibration

The Brier score says *how good*. The calibration plot says *which way to adjust*. **Calibration** is how closely your stated confidences match how often your calls come true: a well-calibrated reader's 70% calls come true about 70% of the time, his 90% calls about 90%. To see it, group your calls by the confidence you stated, and for each group plot the stated confidence against the share that actually came true. Perfect calibration is the diagonal.

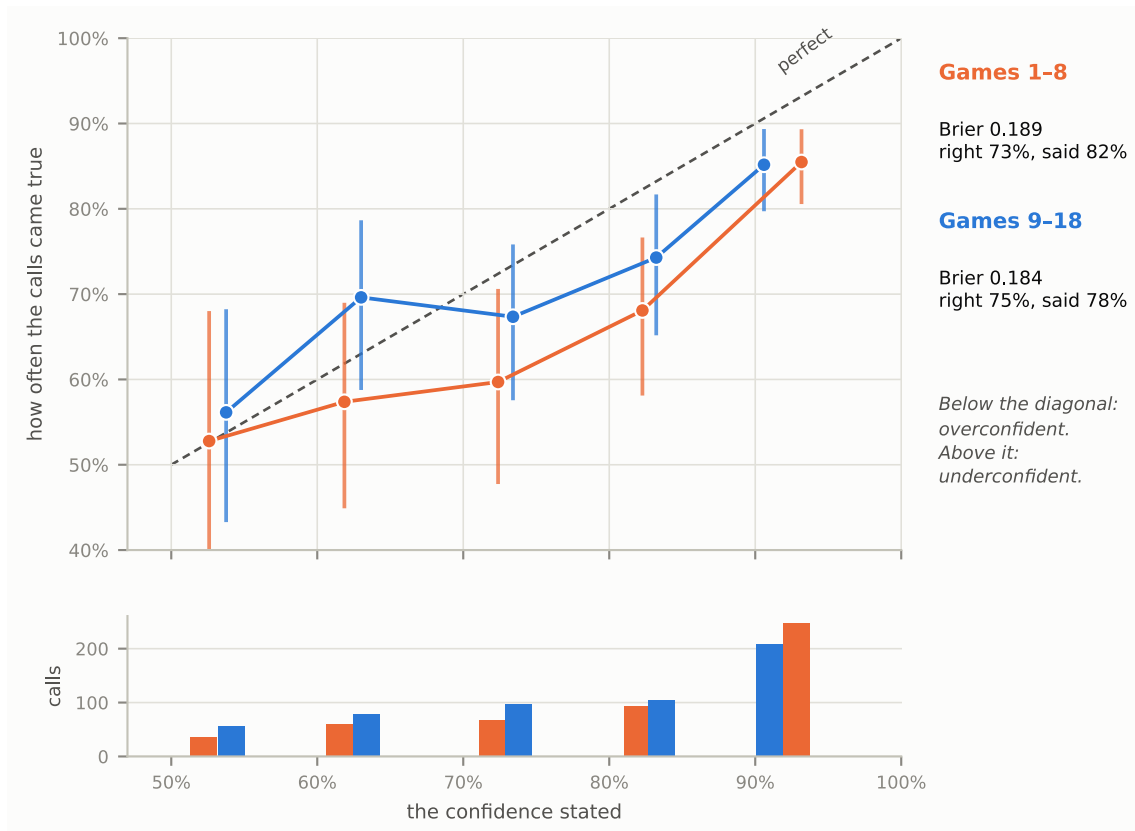


Figure 7: The calibration plot of the example log: every Panthers snap of 2025, called by a simulated reader (real snaps and results, simulated calls). Calls are grouped by the confidence stated (50–55%, 60–65%, and so on); each dot is a group’s actual hit rate, plotted at its average stated confidence, with a 95% interval (the vertical bar) for how much a group of that size could wobble by chance. The diagonal is perfect calibration; below it means overconfident. Orange, games 1 to 8: from 60% up, every group comes true less often than stated, by about ten points in the 70s and 80s. Blue, games 9 to 18: closer to the diagonal through the middle, though still low at the top. The bars at the bottom count the calls in each group: the early reader put about half his calls at 90% or above, the later one spread them out.

Three things to read off a calibration plot, in this order.

Where are the dots relative to the diagonal? In the first half of the season the example reader’s dots sit well below it: his 70–75% calls came true 60% of the time, and his 90%-plus calls 85%. That’s **overconfidence**, the commonest pattern there is: in study after study, people’s “90% sure” comes true well short of 90% of the time (Watching Live cited the classic review). The fix is mechanical. Pull the confidences in: when you feel 90%, write 80% for a few weeks, and watch whether the dots climb toward the line.

How wide are the bars? A dot from 30 calls can sit ten points off the diagonal by chance alone; a dot from 300 can’t. The vertical bars are 95% intervals for that wobble.¹¹ Where a bar crosses the diagonal, the gap may be noise. This is why one game can’t tell you whether you’re calibrated: sixty calls spread over five groups is about a dozen per group, and every bar would span thirty points. A season can.

How spread out are the dots? This is the one people forget. A reader who says 60% on every snap of a team that throws 60% of the time is *perfectly* calibrated: his single dot sits right on the diagonal. He’s also useless; he never tells you anything the team’s pass rate didn’t. What makes a reader good is **resolution**: stating different confidences on different snaps, and having those differences come true. On a calibration plot, resolution is the dots being spread from the bottom-left to the top-right, with the 90% calls really coming true more often than the 60% calls. The Brier score rewards both. Calibration is knowing how sure to be; resolution is knowing more.

¹¹The intervals are Wilson score intervals: Edwin B. Wilson, “Probable inference, the law of succession, and statistical inference,” *Journal of the American Statistical Association* 22 (1927): 209–212. They behave better than the simple “plus or minus two standard errors” when a group is small or its hit rate is near 100%.

i Go deeper: the Brier score in three pieces

Allan Murphy showed in 1973 that the Brier score splits exactly into three parts:¹²

$$\text{Brier} = \underbrace{\text{reliability}}_{\text{calibration error}} - \underbrace{\text{resolution}}_{\text{knowing more}} + \underbrace{\text{uncertainty}}_{\text{the game's own}}$$

Write each call as a chance of a pass (a “run, 70%” call is a 30% chance of a pass) and group the calls by that chance. *Reliability* is the average squared gap between each group’s stated chance and its actual pass rate: calibration error, lower is better. *Resolution* is how far the groups’ actual pass rates spread from the overall pass rate: higher is better, and it’s the only term that rewards reading. *Uncertainty* is $p(1 - p)$ for the overall pass rate p : the part nobody can remove, the same for every reader of the same snaps.

For the example log, games 1 to 8: reliability 0.020, resolution 0.075, uncertainty 0.244. Games 9 to 18: reliability 0.009, resolution 0.064, uncertainty 0.238. The calibration error halved: that’s the reader getting honest, and it’s the cheap kind of improvement, because it takes no new knowledge. Resolution slipped a little, partly because the later games were a different mix of snaps, which is why the total Brier score moved less than the calibration did. Raising resolution takes reading more, which is what the rest of this course was for.

Scoring the log in Python

Here’s the scoring routine. It reads a log in the final form from Watching Live (you need at least the `call`, `conf` and `result` columns, and `clue` and `game` if you have them) and prints everything this section has discussed. It’s run here on the example log, saved as `prediction_log_2025.csv`.

```
import numpy as np
import pandas as pd

def score_log(log, base_rate=None):
    """Hit rate, Brier score, skill against the base rate, and the calibration table of a prediction log."""
    hit = (log["call"] == log["result"]).astype(float)
    conf = log["conf"].astype(float)
    brier = ((conf - hit) ** 2).mean()
    p = (log["result"] == "P").mean() if base_rate is None else base_rate
    ref = p * (1 - p) # Brier score of always saying the base rate
    print(f"{len(log)} calls: right {hit.mean():.1%}, average stated {conf.mean():.1%}")
    print(f"Brier {brier:.3f} (always 50%: 0.250; base rate: {ref:.3f}); "
          f"skill vs base rate {1 - brier / ref:+.2f}")
    groups = pd.cut(conf, [0.49, 0.575, 0.675, 0.775, 0.875, 1.0],
                    labels=["50-55%", "60-65%", "70-75%", "80-85%", "90%+"])
    table = (pd.DataFrame({"stated": conf, "right": hit})
            .groupby(groups, observed=True).agg(calls=("right", "size"), stated=("stated", "mean"),
            right=("right", "mean")))
    table["gap"] = table.right - table.stated # negative = overconfident
    print(table.round(3).to_string())
    return brier

log = pd.read_csv("prediction_log_2025.csv")
score_log(log)
print("\nBy the clue you wrote down:")
log["right"] = log.call == log.result
print(log.groupby("clue").right.agg(calls="size", right="mean").round(3).to_string())
```

¹²Allan H. Murphy, “A new vector partition of the probability score,” *Journal of Applied Meteorology* 12 (1973): 595–600. The split is exact when calls are grouped by their distinct stated values, as here (confidences rounded to 5%).

```
print("\nBrier by quarter of the season:")
log["part"] = pd.cut(log.game, [0, 4, 8, 13, 18], labels=["games 1-4", "5-8", "9-13", "14-18"])
log["sq_gap"] = (log.conf - log.right) ** 2
print(log.groupby("part", observed=True).sq_gap.mean().round(3).to_string())
```

```
1054 calls: right 73.9%, average stated 79.9%
Brier 0.186 (always 50%: 0.250; base rate: 0.241); skill vs base rate +0.23
```

```
      calls  stated  right   gap
conf
50-55%    93   0.532  0.548  0.017
60-65%   140   0.624  0.643  0.019
70-75%   165   0.729  0.642 -0.086
80-85%   199   0.827  0.714 -0.114
90%+    457   0.920  0.853 -0.067
```

By the clue you wrote down:

```
      calls  right
clue
align    561  0.663
sit      493  0.826
```

Brier by quarter of the season:

```
part
games 1-4    0.177
5-8          0.201
9-13         0.189
14-18        0.178
```

The `gap` column is the calibration plot in numbers, and the clue table is the one How to Watch a Broadcast told you to watch: a clue you lean on often that isn't beating the snaps where you only had the situation isn't earning its place. Careful how you read it, though: the two rows aren't the same kind of snap. The `sit` snaps are the ones where the situation alone was lopsided, so of course they come true more often. The fair test of a clue is against a forecaster who can't see it, on the same snaps: on the `align` snaps the example reader was right 66% of the time, and `xpass`, which can't see the formation, 52%. That clue is earning its place.

Reading your report card

Once a month, or at the end of the season, run the score and match what you see to a fix:

What the score shows	What it means	What to change
Dots below the diagonal, worst at the top	Overconfident	Pull every confidence in by about ten points for a month; cap yourself at 85%
Dots above the diagonal	Underconfident: you know more than you admit	Push the confidences out where the formation agrees with the situation
All calls between 50% and 65%	Hedging; low resolution	Use the formation step: Watching Live showed it's the one big move after the situation
Brier worse than the base rate	Your clues are pointing the wrong way	Score by clue; drop or reverse the one that's below 50%
Good on run/pass, poor on the second layer	Normal: coverage is hard	Keep second-layer confidences at 55–70%; check them against your chart
One bad game	Probably nothing	Look at the whole season; a game is too few calls to judge

And keep the earliest sheets. The tally from Downs, Distance, and Field Position had no confidences, so score it by hit rate only and set it against the down-and-distance rule; your first formal log from How to Watch a Broadcast had confidences, so give it a Brier score too. Then put every game since on the same chart:

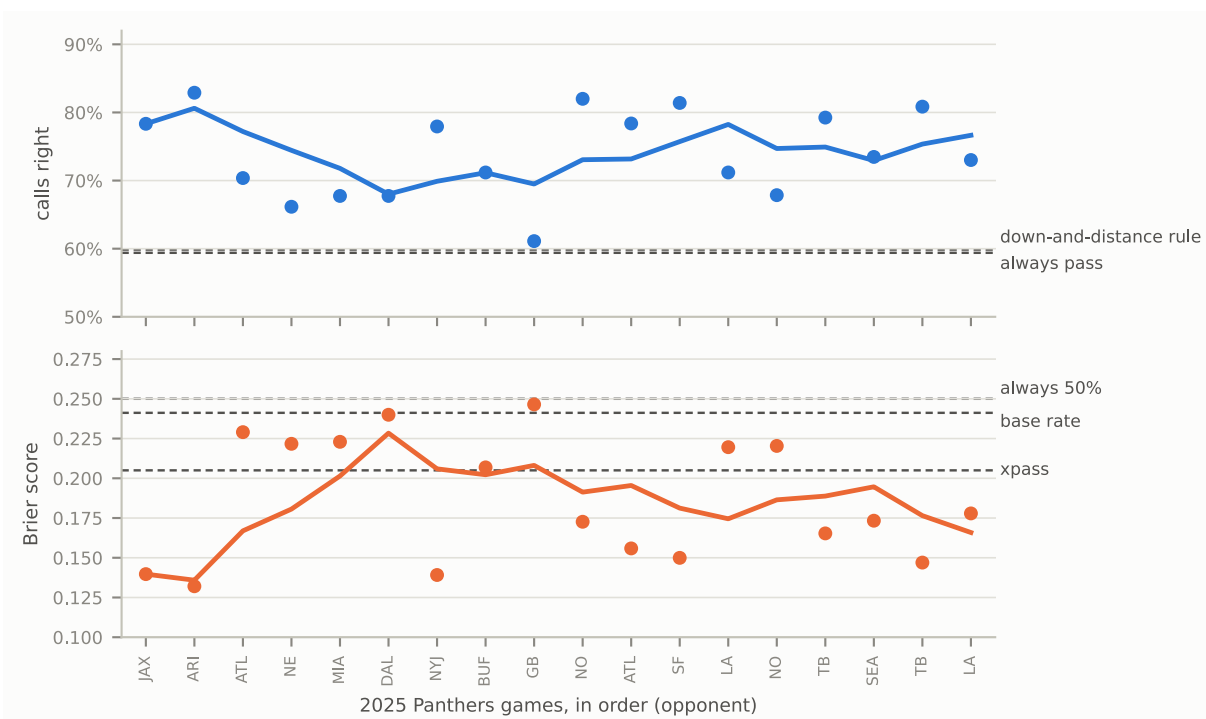


Figure 8: The example reader's season, game by game (every 2025 Panthers snap; real results, simulated calls). Top: the share of calls that came true, with the bars to beat from How to Watch a Broadcast drawn across: 'always pass' and the down-and-distance rule of your first tally, on these same snaps. Bottom: the Brier score per game (lower is better), with the line for saying 50% every time, the base rate, and xpass. The line is a four-game rolling average. The reader beats the scorebug (xpass) in most games, but look how far the dots jump: a game is only 43 to 76 calls, and some games are simply easier to read than others (the first two were).

Week to week, the score mostly measures the games; only across a season does it measure the reader.

Scoring the second layer against your chart

Your run-or-pass calls can be scored from the play-by-play alone. The second layer can't: "zone, 65%" needs to know what coverage was played, "five or more rushers, 60%" needs the rush count, "the run goes left" needs the direction. That's what the chart is for. Join the log to the chart on the snap, and score each second-layer call the same way:

```
# A second-layer call on every New England dropback of Super Bowl LX: "two-high after the snap",
# stated at Seattle's 2025 regular-season rate for the down (the kind of call a scouting card supports)
l2 = SB[(SB.posteam == "NE") & (SB.qb_dropback == 1) & SB.cv.isin(TWO_HIGH | ONE_HIGH)].copy()
l2["l2conf"] = l2.down.map(lambda d: sea_two_rate(int(d))) # the stated chance of two-high
l2["came_true"] = l2.cv.isin(TWO_HIGH).astype(float) # the chart's answer
brier_l2 = ((l2.l2conf - l2.came_true) ** 2).mean()
print(f"{len(l2)} coverage calls; two-high came true on {l2.came_true.mean():.0%}; "
      f"average stated {l2.l2conf.mean():.0%}; Brier {brier_l2:.3f}")
```

```
51 coverage calls; two-high came true on 73%; average stated 53%; Brier 0.221
```

Calls made from Seattle's season-long habit were far too timid for this game: Seattle played two-high on 73% of these dropbacks, against the 52% its regular season suggested. One game can't tell you whether that's a flaw in the calls or a game plan built for one opponent (here, almost certainly the game plan). Over a season you'd score second-layer calls exactly as you score the first layer, with their own calibration plot. Expect it to be flatter: the second layer is harder, and a reader whose coverage calls are calibrated at 60% is doing well.

i Go deeper: Brier score or log loss?

Tendencies scored xpass with *log loss*, the average of $-\ln(\text{probability you gave to what happened})$. It's also proper, and it's what most machine-learning models are trained on. The difference is in how it treats confident misses. A 95% call that misses costs 0.90 in Brier and 3.0 in log loss; a 99% call that misses costs 0.98 in Brier and 4.6 in log loss; a 100% call that misses costs 1 in Brier and infinity in log loss. For a model, that's a useful alarm. For a human log, where one careless 99% would swamp a season, the Brier score's cap is kinder and easier to read, and it's what forecasting tournaments use for people.

Where to keep learning

This is the end of the course, not of the subject. Four directions are worth your time.

The coaches themselves. Coaches teach each other in clinics: talks at the American Football Coaches Association's annual convention, regional clinic tours such as the Nike Coach of the Year and Glazier clinics, and the notes and videos that circulate from them. A clinic talk is a coach explaining one scheme to other coaches, with the drawings, the rules, and the problems it solves; it's the closest thing to the inside of a meeting room a fan can get, and it's where the "why" behind most of this course came from. Read them with the vocabulary you now have, and with the habit from Constraint Theory and Sequencing: every scheme has a weakness the coach giving the talk is managing. Coaches have written good books, too. Bill Walsh's *Finding the Winning Edge* is the classic on running an organisation and a game plan; for the ideas behind modern schemes, Chris B. Brown's *The Art of Smart Football* and Tim Layden's *Blood, Sweat and Chalk*.

The film-breakdown community. A generation of writers and video makers now does on the internet what this chapter asks you to do: All-22 clips, frozen frames, arrows. Ted Nguyen at *The Athletic*, Cody Alexander's MatchQuarters on defense, Brett Kollmann's film-room videos, and the quarterback-focused breakdowns of JT O'Sullivan's QB School are good places to start, and Greg Cosell's decades of All-22 study at NFL Films are behind much of the public vocabulary. Watch how they work: freeze, rewind, one group at a time. Disagree with them when your own film says otherwise.

The analytics community. Everything in Part 13 came from people who publish their data and code for free. The nflverse project (and its Python package, nflreadpy) is the foundation; Open Source Football collects worked analyses with code; Ben Baldwin's writing explains the models behind EPA and xpass; and FTN's charting, free to use with credit under its CC-BY-SA licence, is the public coverage and alignment data this course leaned on.

The Big Data Bowl. Every year since 2019 the NFL has released a slice of its player-tracking data and asked the public to do something new with it, and its best entries have changed how the league measures the game (Tracking Data and the Big Data Bowl). The 2026 edition asked about player movement while the ball is in the air; as of early October 2026 the 2027 edition had not been announced, and past editions have opened in the fall.¹³ If you've done the course, you can enter it: you can read a play, chart one and model one, and the `gridiron` library you've been reading diagrams from takes Big Data Bowl tracking files directly.

i Go deeper: a weekly routine

A season-long loop that fits around a job:

- **Sunday:** the live log for one team, every snap, with a confidence (one half if that's all you can manage).
- **Monday or Tuesday:** the film for that game. Step 0 and pass 3 on every snap; the full protocol on the 20 to 30 snaps that decided it. Fill the chart.
- **Midweek:** score the log, update your one-page card for the team (Tendencies) from your chart, and pick one thing to watch for next Sunday.
- **Once a month:** the calibration plot, and the report card above.
- **In the off-season:** one project with tracking data, the Big Data Bowl if it's running.

¹³NFL Football Operations, "Big Data Bowl": <https://operations.nfl.com/gameday/analytics/big-data-bowl>. The 2026 edition (the eighth) studied player movement while the ball was in the air, using 2023-season tracking data; no 2027 edition had been announced as of October 6, 2026.

💡 Watch for it on Sunday: chart a full game

One game, one team's offense against the other's defense, end to end:

- **Before kickoff**, build the skeleton with `chart_skeleton()` from last week's game to test your setup, and write your codebook rules (the strength rule, where the box ends) at the top of a blank sheet.
- **During the game**, keep the live log exactly as in Watching Live: every snap, the checklist, a call and a confidence before the snap, and a second-layer call when there's time.
- **The day after**, build the real skeleton for this game, and with the coaches film run step 0 and pass 3 on every snap and the full four-pass protocol on third downs, red-zone snaps, sacks, turnovers and explosives. Fill every film column; use ? without shame.
- **Then score it**. Run `score_log()` on the game's log for the first layer, and join the log to your chart to score the second layer. Write down the three snaps that cost you the most Brier score, and the "because" line for each.
- **Keep the files**. One game is a start. A season of them is a calibration plot, a chart of a team, and the best football education you can give yourself.

Predict the play

These drills are the film room's version of the course's habit: predict first, then check. The first two are pre-snap frames from the film, and the question is what the next pass of the protocol will show. The third is a scoring drill.

💡 Predict the play 1: pass 3, before you press play

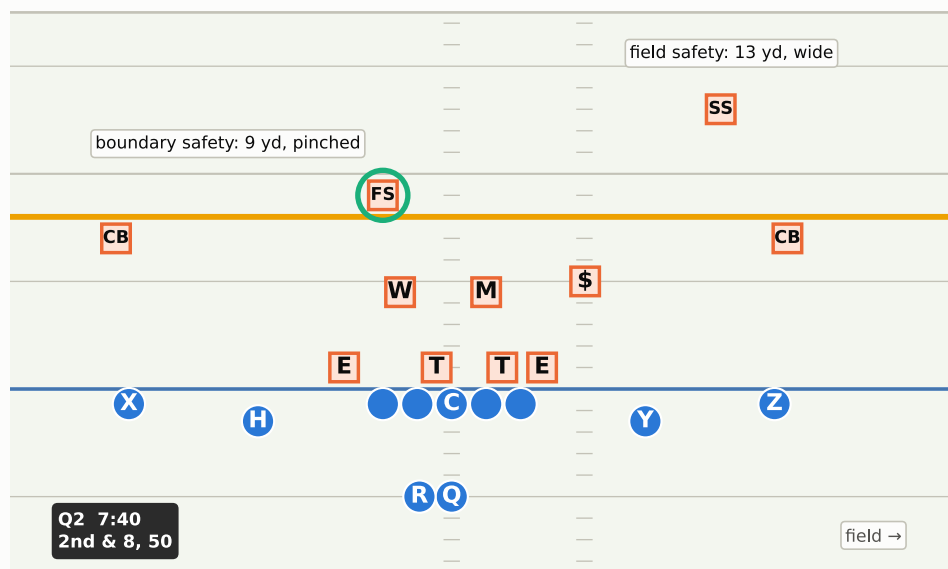


Figure 9: Frozen at the snap. Second-and-8 at midfield, ball on the offense's left hash, so the wide side of the field is to the right. Shotgun, 11 personnel, two receivers each side. The defense is in nickel with two safeties deep, but not evenly: the safety on the short side (left, ringed) is at about 9 yards and pinched toward the ball, while the field safety (right) is at 13 yards and wide. Both corners are about seven yards off with outside leverage, and the nickel (\$) is splitting the difference between the slot and the tackle.

What coverage will pass 3 show after the snap, how sure are you, and what will you watch first to find out?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: pass 1, a guard on his heels

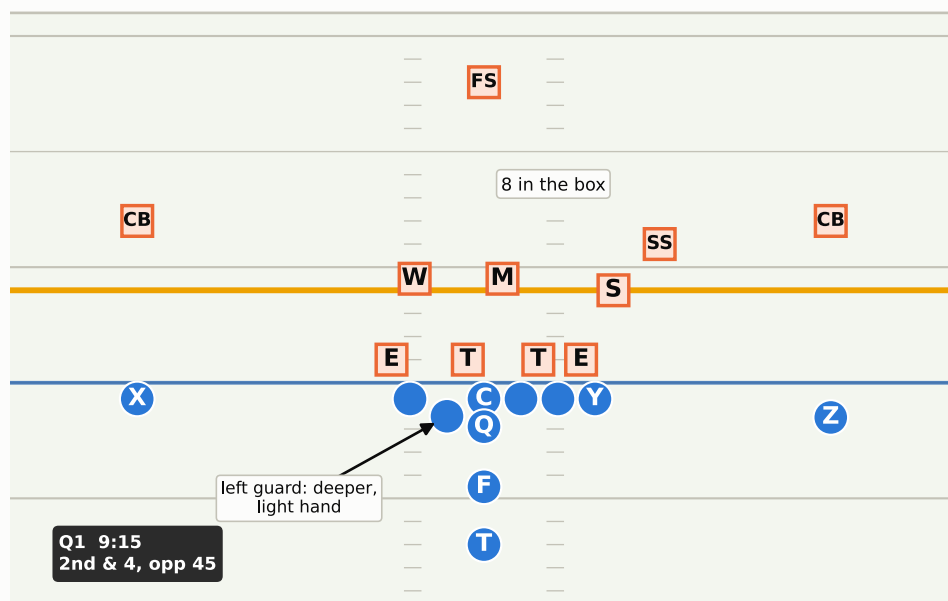


Figure 10: Frozen at the snap. Second-and-4 at the opponent's 45, first quarter. 21 personnel in an I-formation: the quarterback under center, the fullback (F) and tailback (T) behind him, the tight end (Y) on the right. The defense is in base with one safety deep and the strong safety walked down: eight in the box. On the end-zone angle, the left guard (arrow) is set a little deeper than the center and the right guard, with less weight on his hand (exaggerated here so it shows).

What will pass 1 show at the snap, which way will the run go, and which defender will nobody block?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: two readers, one season

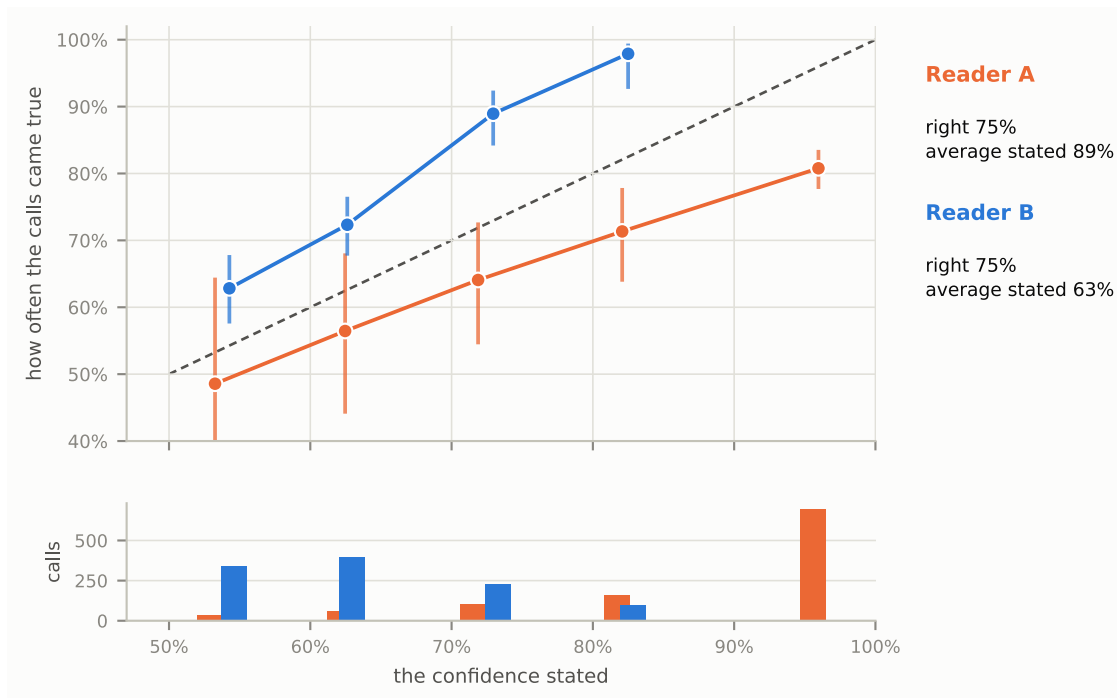


Figure 11: Two readers' calibration plots for the same season of snaps (every 2025 Panthers snap; real results, simulated calls). Both called run or pass on every snap and got almost exactly the same share right. Reader A (orange) stated 90% or more on most calls. Reader B (blue) never went above 85%, and most of his calls were 50–65%. The bars at the bottom count each reader's calls at each confidence level.

Which reader has the better Brier score? What's each one's mistake, and what should each change?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Film answers what Sunday can't.** The All-22 shows all 22 players for the whole play; the coaches film's end-zone angle shows the line. Use them to check your reads, not to rewrite them.
- **Four passes, in order.** Freeze at the snap and grade your read; then the line, the quarterback, the coverage, the skill players, from what sets the clock to what uses it; then one "because" line that says whose play it was.
- **A chart turns a game into rows.** Same fields, same codes, every snap, with ? allowed. Let the play-by-play fill a third of it; the film columns are where your chart knows more than any public file.
- **Score with a proper rule.** The Brier score rewards being right and being honest about how sure you were; 0.25 is "I have nothing", and the base rate and xpass are the bars to beat.
- **Calibration is how sure; resolution is how much.** Read the calibration plot for which way to adjust, the spread of the dots for whether you're learning anything, and never judge either from one game.

Connections

Builds on: the master checklist, stated confidence and the final form of the log from Watching Live; the base rates, xpass, coverage charting and the caveats of charted labels from Tendencies; coaches film, the camera angles and the first prediction log from How to Watch a Broadcast; your very first tally in Downs, Distance, and Field Position; and the QC coaches' tags and tendency report from The Game Plan.

Sets up: everything after the course. The Big Data Bowl and the tracking data of Tracking Data and the Big Data Bowl are the natural next project; the atlases (appendices A to D) are the reference cards to keep beside the film.

Case studies: the film rooms here were Seattle's defense in Super Bowl LX, the end point of Fangio's Two-High Revolution and Macdonald's Answer.

i Go deeper

- Philip Tetlock and Dan Gardner, *Superforecasting* (2015): the forecasting tournaments that scored people with the Brier score, and the habits of the ones who did best.¹⁴
- Nate Silver, *The Signal and the Noise* (2012): the chapter on weather forecasting is the best plain-English account of calibration, and of why the forecasters who are scored on it get good at it.¹⁵
- Bill Walsh with Brian Billick and James Peterson, *Finding the Winning Edge* (1997), and Chris B. Brown, *The Art of Smart Football* (2015): how coaches think about a game plan and its film.
- The nflverse data dictionaries (play-by-play, participation, FTN charting) and Open Source Football: the columns behind your skeleton, and worked analyses to copy.
- The NFL's Big Data Bowl page at operations.nfl.com and the competition pages on Kaggle: past winners' notebooks are a free course in tracking data.

Answers to Predict the play

Answer 1.

Single-high after the snap, about 60–65%, most likely Cover 3 with the boundary safety rotating down; watch the field safety's first step. The shell is two-high, but it isn't a balanced two-high: one safety is four yards shallower than the other and pinched toward the ball. That's the tell from the tells catalogue in Watching Live: when two safeties disagree, the shallower, narrower one is usually the rotator. He's already close enough to drop into a short zone on the boundary side at the snap (a buzz rotation if he goes to the hook area, a sky rotation if he goes to the flat), and the field safety, already wide, would then run to the deep middle. The corners agree: off and outside leverage is how corners line up to bail into deep thirds with help inside (The Cover 3 Family).

Why not 90%? Because disguise is the norm (Disguise, Rotation, and Split-Field Coverage): the pinched safety could be a quarters safety cheating to take away a slot route, or bait for a quarterback who reads exactly this tell. And Cover 1 is possible too, if the corners turn and run with their receivers instead of bailing.

What to watch: the **field safety's first step**. If he drives toward the middle of the field, the middle is closing and it's single-high. If he backpedals straight, it's still two-high, and the pinched safety was playing a quarter or a half from a cheated spot. Then the corners: bail (Cover 3) or turn and trail (Cover 1). On the chart: **shell 2**, then **cov C3**, **rot B** if you were right.

Answer 2.

A gap run to the right, most likely power: the left guard pulls, the line blocks down, and the backside end is left alone. Run about 75%; right, about 65%. Start with run or pass: second-and-4 from 21 personnel under center is a run look, though eight in the box is the defense inviting play-action, so not 90%.

The guard is the scheme tell from Watching Live's catalogue: a lineman about to pull sideways sets up for it, with less weight forward and a little depth. A pulling left guard runs to the right, toward the tight end, so this looks like **power** (Gap and Power Runs). What pass 1 should show, if that's right: at the snap the center, right guard, right tackle and tight end all block *down*, toward the left; the fullback runs at the defender on the end of the line on the right and kicks him out; and the left guard pulls behind the line and turns up through the hole the kick-out and the down blocks create, looking for a linebacker.

Who's unblocked? The defensive end on the *left*, the backside. The left tackle steps inside to fill the gap the guard left, and the end outside him is too far from the play to matter, if the play is fast enough. That's the trade power makes: it spends the backside guard to get an extra blocker at the point of attack, and it accepts one free defender chasing from behind. If the film shows that end making the tackle, the "because" line is about speed: the play was too slow, or the back hesitated.

And if the line instead steps *together*, all to one side, with no puller, the guard's stance was a false tell: it's zone, and you note it in your clue column.

Answer 3.

¹⁴In the Good Judgment Project, the team Tetlock led in the forecasting tournament run by IARPA (the U.S. intelligence community's research agency) from 2011, every forecaster was scored with the Brier score; Tetlock and Gardner, *Superforecasting: The Art and Science of Prediction* (Crown, 2015), ch. 3.

¹⁵Nate Silver, *The Signal and the Noise* (Penguin, 2012), ch. 4, on the calibration of U.S. National Weather Service precipitation forecasts. The original evidence: Allan H. Murphy and Robert L. Winkler, "Reliability of subjective probability forecasts of precipitation and temperature," *Journal of the Royal Statistical Society, Series C (Applied Statistics)* 26 (1977): 41–47.

Reader B scores better (0.187 against 0.195), though both are making a fixable mistake in opposite directions. Their hit rates are almost identical (75% and 75%), so the whole difference is in the confidence column.

Reader A is overconfident. His dots sit below the diagonal all the way up, and his 90%-plus calls, the bulk of his log, came true only 81% of the time. Each of those misses costs between 0.8 and 1 in Brier score. The fix: pull everything in. His calls already sort snaps well (his dots climb from left to right, which is resolution); his numbers are just too big.

Reader B is underconfident. His dots sit *above* the diagonal: his 70% calls came true far more often than 70%. He knows more than he admits, and he never stated more than 85% on any snap. The fix: push the confidences out where the clues agree, toward 90% on the empty-backfield, third-and-long snaps.

How much is at stake? If each reader replaced every stated confidence with the rate at which those calls actually came true (perfect calibration, the same calls), A's Brier score would fall to 0.173 and B's to 0.172. Same reading, better numbers. That is the cheapest improvement in the whole course, and the calibration plot is how you find it.